Reaching the Actionable Radius: Physics-Guided Calibrated Localization of Methane Sources under Wind Uncertainty

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***Abstract*—Inexpensive sensor networks can watch a natural-gas site around the clock, and machine learning can turn their readings into a leak search region in milliseconds, with a statistical guarantee that the region contains the source 90% of the time. We show that the guarantee alone is not enough, and we provide a simple, verified way to make the regions small enough to act on, even when the wind measurement is imperfect. On a simulation testbed whose physics is exactly solvable, we compute the smallest region the sensor data truly support and use it as a measuring stick: a guaranteed deep-learning localizer produced 94–98 m regions where the data supported 6–7 m. We close half of that gap with one change. For every map cell, we compute how well a leak at that cell would explain the readings, a classical signal-processing operation, and we provide these maps to the network as additional input images. With ordinary training and 0.2% added parameters, three different network families all reach 46–48 m, the scale that U.S. EPA super-emitter notifications use to associate a leak with a facility (50 m). We then make the test realistic: the network only ever sees a wind *measurement* with typical instrument error (10° direction, 10% speed). The same recipe, extended so that the input maps also carry the wind sensor's stated error range, retains its guarantee and reaches 70–73 m, with a quarter of cases below 50 m. All results are verified against exact references and reported over multiple training seeds.**

***Index Terms*—methane monitoring, conformal prediction, sensor networks, wind uncertainty, physics-guided machine learning**

I. Introduction

Methane traps roughly 80× more heat than CO_2 over its first 20 years and fades from the atmosphere within about a decade, so stopping large leaks quickly is one of the highest-leverage climate actions available today. A large share of oil-and-gas methane comes from a small number of large point sources, known as super-emitters and defined by the U.S. EPA as events of $100\,\mathrm{kg\,h^{-1}}$ or more [2]. Detection technology has advanced rapidly; UNEP's satellite alert system issued over a thousand notifications in its first two years [1]. The bottleneck is what happens next: connecting an alert to a specific place that a crew can inspect. EPA's Super-Emitter Program (implementation extended to January 2027) asks notifications to identify any facility within **50 m** of the estimated leak location [2], and the EU imposes analogous source-level duties [3]. We therefore adopt 50 m as an operationally motivated target: a region small enough to trigger a focused inspection.

Continuous monitoring networks, in which a small number of inexpensive sensor masts capture the plume as the wind sweeps it across the site, are where this localization can happen. Machine learning provides instant estimates, and a standard statistical wrapper called split conformal prediction [4] adds a guarantee: a region that provably contains the true source 90% of the time. The gap we address is twofold. First, the guarantee speaks only to how *often* the region contains the source, not to how *small* the region is, and there was previously no practical way to measure how much smaller a guaranteed region could have been. Second, the classical alternative of inverting the plume physics directly requires the wind to be known exactly, while every real site has only a wind *measurement*; how each approach behaves under that everyday mismatch had not been carefully quantified.

Contributions.

- **C1: A measuring stick for guaranteed regions.** A methane testbed (CH4-T) whose physics is exactly solvable, so the smallest region the data support is computable for every scenario. This allows any guaranteed localizer to be audited in meters. A verified-90% deep localizer delivered 94–98 m where the data supported 6–7 m.
- **C2: A diagnosis, and a simple fix through the inputs.** Controlled comparisons across three network families and a modern generative estimator point to one missing ingredient: the classical computation that scores, for every map cell, how well a leak there would explain the readings. Supplied as two additional input images (closed form, milliseconds) with a near-zero-cost add-on head and ordinary training, it brings all three families to 46–48 m.
- **C3: One recipe that carries its wind-error budget.** The same input maps, extended with the wind sensor's stated error range (the average evidence, its variability, and a fit-quality map across plausible winds), retain the guarantee under realistic wind error and reach 70–73 m. When the error budget is set to zero, the extended maps reduce exactly to the classical ones, so a single recipe covers both settings.

Every component is standard and auditable: classical detection theory, ordinary supervised training, and textbook conformal

Figure 1 placeholder: wind-swept plume, masts, and true source on the 500 m site, with median 90% regions drawn to scale: smallest supported region (∼7 m), our recipe with exact wind (46–48 m), baseline network (94 m), our recipe with measured wind (70–73 m), and the 50 m facility-association radius.

Fig. 1. The opportunity, drawn to scale (placeholder).

calibration. Every number below comes from one shared, verified simulator, identical data splits, and an identical calibration harness, reported over three training seeds.

II. Related Work

Conformal prediction gives distribution-free coverage guarantees [4]; how small the guaranteed sets are is the open half of the problem, and our audit scores it directly against the exact probability map at matched, verified coverage. **Simulation-based inference** trains neural networks to approximate probability maps from simulated data [6], [7], [8]; its standard diagnostics check calibration [5], and recent work examines what happens when the simulator's settings differ from reality at deployment [9]. Our measured-wind design addresses exactly that setting by building the wind sensor's declared error range into the network's inputs. Our maps are classical statistics chosen by hand, transparent and checkable line by line, which complements approaches that learn their summaries from data. **Source-term estimation** inverts dispersion physics with optimization or sampling at minutes to hours per event [10]; learned localizers answer in milliseconds, and our audit adds the missing instrument for judging how much information they extract. **Controlled-release testing** (METEC) evaluates continuous monitoring systems in the field [11]; our testbed mirrors that task structure with exact ground truth, complementing rather than replacing field validation.

III. Methodology

A. Testbed, measured wind, and the audit (C1)

CH4-T places one source of unknown location and rate (10–$500\,\mathrm{kg\,h^{-1}}$, spanning the $100\,\mathrm{kg\,h^{-1}}$ super-emitter threshold) on a $500\times500\,\mathrm{m}$ site watched by 4–12 masts (ppm readings with realistic noise and dropout). Gas transport follows the standard Gaussian plume model with published Pasquill–Gifford/Briggs dispersion coefficients (verified against the original tables; mass conservation checked to 4×10^{-6}), driven by a naturally meandering wind record (1–$8\,\mathrm{m\,s^{-1}}$; 1-minute steps over a 30-minute window). **Wind uncertainty is built into the data.** Readings are generated under the true wind, but the dataset records only the *measured* wind: the true wind corrupted by 10° of direction error and 10% of speed error per step, the accuracy class of inexpensive monitoring anemometers. No model, at any stage, observes the true wind. Because the plume model is closed-form and linear in the leak rate, the exact probability map over a 64×64 grid is computable for every scenario under known wind (high-precision quadrature; normalization checked to $\sim10^{-12}$). This provides the measuring stick: the smallest 90% region the data support is 7.0 m at nominal settings and 6.2 m when coverage is matched to the learned models' 0.89. A blocking check requires the calibration wrapper, applied to the exact map itself, to reproduce its own guarantee before any experiment proceeds; this check caught two real implementation bugs, including a tie-handling correction that we report for reuse. All models share identical data (10k/1k/2k/2k splits, a standard budget for this field [8]), identical symmetry augmentation, and an identical conformal wrapper, always calibrated under the same wind condition in which it is evaluated. Region sizes are reported as the radius of the circle of equal area.

B. Diagnosis: what was missing (C2)

A standard set-based network trained on ordinary labels is correctly calibrated (coverage 0.89) at 94–98 m, so the information required for 6–7 m is present in the readings and waits to be extracted. Three controlled probes narrow down where it goes missing. A modern generative head (a flow-based neural posterior estimator, strengthened with more training and a wider architecture after review of our own implementation) reaches 62.8–71.2 m; since we verified that its output layer can express 4.4 m regions, expressiveness is not the limiting factor. The classical physics score by itself, calibrated with no network at all, returns essentially the whole site: it identifies where the evidence lies but not how much to trust it. What closes the gap on every network we tested is supplying the computation itself as input: for each of the 4,096 candidate cells, correlate the readings against the pattern that a leak at that cell would produce. Networks calibrate excellently and compute this poorly; classical physics computes it exactly and calibrates poorly. The method is the combination.

C. One recipe, carrying its wind-error budget (C3)

For any base localizer (Fig. 2), let g_c be the pattern of readings that a unit-rate leak at cell c would produce under the *measured* wind, and let $a_c{=}\sum g_c y$ and $b_c{=}\sum g_c^2$ summarize how the actual readings y align with that pattern. **(1)** State the wind sensor's error range ε (from its specification sheet), draw eight plausible true winds within that range, and compute four 64×64 input images: the *average evidence* $\bar{z}_c$ (how strongly the readings point to cell c, averaged over plausible winds), its *variability* across those winds (how fragile that evidence is to wind error), the average *sensitivity* $\overline{\log b_c}$ (how visible a leak at c would be to this sensor layout), and the average *fit quality* $\bar{R}_c$, the residual error that remains after the best leak rate at c attempts to explain the readings, averaged over plausible winds. Every image is closed-form linear algebra, computed in milliseconds from the network's own inputs. **(2)** Add a small convolutional head ($\sim$10k parameters, 0.2% of the model)

TABLE I

THE AUDIT WITH EXACT WIND: MEDIAN 90% REGION RADIUS AGAINST THE SMALLEST SUPPORTED REGION; 2,000 TEST SCENARIOS.

Method	Coverage	Radius (m)
Smallest supported region (exact physics)	0.89–0.95	**6.2–7.0**
Deep localizer, labels only	0.89	94–98
Ours (zero-budget setting)	0.89	**47.1–47.7**

TABLE II

EXACT WIND; 3 SEEDS PER CELL; COVERAGE 0.88–0.92 THROUGHOUT.

Estimator	Without maps	With maps
DeepSets	94–98	**47.1–47.7**
GNN	55–58	**46.2–47.5**
Set Transformer	69–75	**46.2–47.3**
Flow-based estimator (strengthened)	62.8–71.2	57.6–61.2
Classical score alone (no network)	—	282 (full site)

initialized at exactly zero, so that at the first training step the model is the unmodified baseline and every comparison is controlled by construction. **(3)** Train with ordinary supervised labels. **(4)** Calibrate with the standard conformal wrapper under the deployment wind condition. **When the error budget is zero, the recipe reduces exactly to the classical two-channel form.** With only one wind, the variability image is identically zero, and the fit-quality image becomes a fixed transformation of the evidence image ($R_c = yy - \sigma^2 \max(z_c, 0)^2$), so the four images carry the same information as the classical pair ($z_c, \log b_c$). One recipe covers both settings; the exact-wind results below are its zero-budget instantiation.

Why not invert the physics directly? On this testbed one can; that is our measuring stick. In deployment, realistic dispersion has no exact solution and physics inversion takes minutes to hours per event [10] where monitoring needs seconds, and as §IV-C shows, the learned recipe holds up under measured wind in a setting where direct inversion returns regions too large to act on.

IV. RESULTS

A. The audit: a guarantee is not yet a useful region

Both learned rows pass every standard calibration check; only the audit reveals that one of them delivers regions more than ten times larger than the data support (Table I). Two input images and a small head close half of the distance: the median radius falls by 50.3% (bootstrap 95% CI [48.9, 52.0]), 96.4% of individual scenarios improve, and 51% of regions fall below the 50 m facility scale (percentiles p50/p75/p90 of 49/112/170 m). The median case reaches the scale, and the upper tail marks the clearest room for future work.

B. The same fix works on every estimator we tested

Without the maps, the choice of network changes results by 40 m; with them, all nine runs of three network families land inside a 1.5 m band (Table II). The gain is available to whatever architecture a practitioner already has, with ordinary training and near-zero added cost. The maps also transfer onto the generative estimator and improve it as well; the recipe therefore composes with other estimators rather than competing with them.

TABLE III

MEASURED WIND (10°/10% PER-STEP ERROR; NO METHOD SEES THE TRUE WIND; CALIBRATION UNDER THE DEPLOYMENT CONDITION; 3 SEEDS FOR LEARNED ROWS). MIDDLE ROWS USE PROGRESSIVELY LESS OF THE ERROR-RANGE INFORMATION.

Pipeline	Cov.	Radius (m)	$<$50 m
Direct physics inversion	1.00	282 (full site)	0%
with tuned extra noise allowance	1.00	282 (full site)	0%
Network, no physics images	0.90–0.91	107.9–110.5	0%
Ours, single-wind images	0.91–0.92	82.2–84.2	15–17%
Ours, + average and variability	0.89–0.91	74.7–76.3	20–21%
Ours, + fit-quality image	0.90–0.92	**70.1–72.9**	**24–27%**

C. Measured wind: the main result

Real anemometers have error, and the effect on direct physics inversion is dramatic. Trusting the measured wind exactly, it concentrates its probability so far from the true source that, to keep its guarantee, the calibration wrapper must return essentially the entire site (Table III); allowing extra measurement noise, tuned on held-out data, does not change this. The learned recipe behaves differently, and the pattern is clean: the more of the wind sensor's error-range information the input images carry, the smaller the guaranteed regions become, from 82–84 m using the measured wind alone, to 74–76 m after adding the average and variability of the evidence across plausible winds, to **70–73 m** after adding the fit-quality image, with a quarter of cases below 50 m. This is the same network, the same training, the same guarantee, and one forward pass. The mechanism is constructive: because the physics and the sensor's own error range live in the inputs, ordinary supervised training gives the network thousands of examples from which to learn how the error behaves and how much to trust each cell's evidence. A useful summary of the whole study: 94→47 m came from supplying a missing computation; 47→71 m is the measured cost of imperfect wind at this error range; and 108→71 m is the total value of physics in the inputs when the wind is realistic.

V. LIMITATIONS

CH4-T is a measuring instrument, not a deployment: single steady sources (no zero-leak or multi-source scenarios), flat terrain, known stability class, and a 64×64 grid whose ∼7.8 m cells sit near the reference scale (a grid-refinement study is planned). The 6–7 m reference is a property of the testbed's assumptions, not a physical limit, and it exists only under known wind; the corresponding limit under measured wind is not yet computable, so we report our 70–73 m against the 50 m operational scale rather than an information bound. The wind error range is taken as known from the instrument specification; quantifying sensitivity when the specification is wrong is a natural next step. With exact wind, half of individual cases still exceed 50 m (p90 = 170 m), and the largest leaks

Figure 2 placeholder (full width): unified pipeline. Readings + measured wind + stated error range ε $\rightarrow$ four physics input images (average evidence, variability, sensitivity, fit quality; closed form, ms; at ε=0 these reduce to the classical pair) $\rightarrow$ any localizer + zero-initialized convolutional head, joined at the output $\rightarrow$ guaranteed 90% region calibrated under the deployment condition (46–48 m with exact wind; 70–73 m at 10°/10%; 50 m facility scale for reference).

Fig. 2. One budget-aware, physics-guided, guaranteed localizer (placeholder).

Figure 3 placeholder: measured-wind results as a dot-range plot of Table III (radius vs. configuration, 3-seed ranges), with the full-site inversion bar and the 50 m line for scale.

Fig. 3. Guaranteed region size vs. wind-error information in the inputs (placeholder).

TABLE IV
COMPONENT CHECKS (EXACT WIND), ALL LABEL-TRAINED.

Configuration	Radius (m)	Coverage
Labels only (baseline)	94–98^{3s}	0.89
Maps + head (**the recipe**)	**47.1–47.7**^{3s}	0.89
evidence image only	65.0	0.88
sensitivity image only	98.5	0.90
Classical score only (no network)	282 (full site)	1.00
Smallest supported region	6.2–7.0	0.89–0.95

localize somewhat worse; these are clear, well-defined targets for follow-up work. Field validation on controlled releases is the natural next stage, and the comparisons reported here are exact regardless, because every method shares one verified simulator, one dataset, and one calibration harness.

VI. Conclusions

On a testbed with exact ground truth, we measured what a coverage guarantee is worth and what it still leaves unrealized: a calibrated deep localizer delivered a tenth of the sharpness its data supported. Supplying one classical computation as input images brought three network families to 46–48 m at the 50 m facility-association scale, and the same recipe, carrying the wind sensor's stated error range in its inputs, retained its guarantee at 70–73 m under realistic wind error, with millisecond inference, ordinary training, and 0.2% added parameters.

The research impact is a transferable evidence standard: exactly solvable testbeds as measuring instruments for learned inference, self-checking calibration gates, and one-change-at-a-time controls, applicable wherever guaranteed regions are produced, from gas leaks to acoustic or radiological source search. The social impact is practical: regions small enough to act on, produced in milliseconds from inexpensive hardware, with stated guarantees, and with the wind sensor's imperfection budgeted honestly. This is the missing analytical step between a satellite alert and a crew knowing where to look.

Appendix

Appendix: Ablation and Control Record

DeepSets encoder, identical data and calibration harness, median 90% region radius (m); ^{3s} marks 3-seed ranges.

Reading. Both images contribute, and the evidence image (the correlation computation) carries most of the gain; the sensitivity image alone, which never touches the readings, adds little, which confirms that the correlation computation is the operative ingredient. The classical score without a network pays for its overconfidence with full-site regions, so the learned calibration is load-bearing. By leak rate, the recipe's median radius and fraction below 50 m are: 10–30 $\mathrm{kg\,h^{-1}}$: 44.8 m, 54%; 30–100: 44.5 m, 55%; 100–250: 50.3 m, 49%; 250–500: 55.2 m, 41%.

References

[1] UN Environment Programme (IMEO), "An eye on methane: Methane Alert and Response System annual report," 2024.

[2] U.S. EPA, "Standards of performance for GHG emissions: NSPS OOOOb/EG OOOOc and the Super-Emitter Program," 89 FR 16820, 2024; implementation timing per the 2025 Interim Final Rule.

[3] European Union, "Regulation (EU) 2024/1787 on methane emissions reductions in the energy sector," OJ L, 2024.

[4] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, 16(4), 2023.

[5] S. Talts *et al.*, "Validating Bayesian inference algorithms with simulation-based calibration," arXiv:1804.06788, 2018.

[6] G. Papamakarios and I. Murray, "Fast ε-free inference of simulation models with Bayesian conditional density estimation," *NeurIPS*, 2016.

[7] C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios, "Neural spline flows," *NeurIPS*, 2019.

[8] J.-M. Lueckmann *et al.*, "Benchmarking simulation-based inference," *AISTATS*, 2021.

[9] P. Cannon, D. Ward, and S. M. Schmon, "Investigating the impact of model misspecification in neural simulation-based inference," arXiv:2209.01845, 2022.

[10] T. Newman *et al.*, "Probabilistic inversion modeling of gas emissions: MCMC estimation of Gaussian plume parameters," arXiv:2408.01298, 2024.

[11] C. Ilonze *et al.*, "Assessing the performance of emerging and existing continuous monitoring solutions under a single-blind controlled testing protocol," *Elementa*, 2025.